PORTFOLIO PROJECT · OPTIMIZATION

A third off distribution costs

Linear programming for a luxury retailer's shipping network across India, solved in Excel Solver

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3 warehouses · 6 boutiques · 18 routes

THE NETWORK

490 units of supply, 430 of demand. Who ships where?

Warehouse	Capacity / month
Mumbai	180
Delhi	150
Bangalore	160
Total supply	490

Boutique	Demand / month
Delhi CP	90
Jaipur	60
Hyderabad	70
Chennai	80
Kolkata	55
Pune	75

THE MODEL

A linear program with 18 decisions and 9 rules

Decide

Units shipped on each of the 18 warehouse-to-boutique routes.

Minimize

Total cost: units on each route times that route's freight rate.

Subject to

Each boutique gets exactly its demand. No warehouse exceeds capacity. No negative shipments.

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minimize   Σ cost[w,b] × units[w,b]
subject to Σ_w units[w,b] = demand[b]   for each of 6 boutiques
           Σ_b units[w,b] ≤ capacity[w] for each of 3 warehouses
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THE RESULT

Same warehouses, same boutiques, a third less spent on freight

32.9%

lower monthly cost
₹2,03,800 down to ₹1,36,750

₹8 lakh

saved per year
at today's demand

THE OPTIMAL PLAN

Seven routes, every boutique fully served

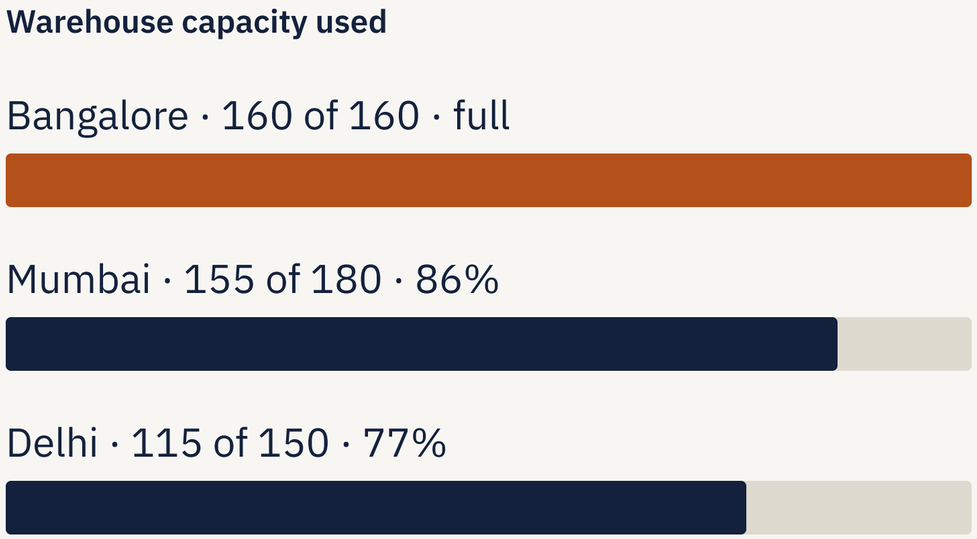
From	Delhi CP	Jaipur	Hyderabad	Chennai	Kolkata	Pune	Used / capacity
Mumbai	90	—	—	—	—	65	155 / 180
Delhi	—	60	—	—	55	—	115 / 150
Bangalore	—	—	70	80	—	10	160 / 160
Demand met	90	60	70	80	55	75	430 / 430

Units shipped per month. A dash means the route is unused.

THE KEY INSIGHT

The cheapest route is not always the cheapest plan

Bangalore is the cheapest source for Pune: ₹240 a unit against ₹290 from Mumbai. But Hyderabad and Chennai have no good alternative, since their next-best warehouse costs ₹590 and ₹670. Bangalore serves them first, and Pune gets only its last 10 units.



SENSITIVITY AND NEXT STEPS

Where the plan is sensitive

Scenario	Monthly cost	What changes
Bangalore capacity +20 units	−₹1,000	20 more Pune units shift to Bangalore
Mumbai capacity +20 units	₹0	Nothing: Mumbai already has slack
Kolkata demand +25 units	+₹13,500	Delhi absorbs it at ₹540 a unit
Bangalore to Chennai freight +15%	+₹3,760	Nothing: routes stay the same

Recommendation: adopt the plan as the monthly shipping policy and re-solve it each month as demand moves. Skip any Mumbai expansion.